

Email #1: Email to a new data professional on the NPS data team

Dear Akbar,

My name is Aquib, one of the data professionals with the National Park Service and a member of the data team responsible for the visitation prediction project.

Welcome to the National Park Service data team! We are thrilled to have you on board. I am reaching out to give you a comprehensive overview of our upcoming project, outline our team's workflow, and align on our high-priority technical goals so you can hit the ground running.

Project Objective & Background Our primary objective is to build a machine learning model to predict future visitation at the most heavily trafficked National Parks. To give you some context, the NPS saw a massive spike of 312 million recreation visits in 2022 (a 15-million-visit increase from 2021). Crucially, **26% of that total volume is concentrated in just the top 8 most visited parks** (such as Yellowstone, Yosemite, and the Grand Canyon). This surge is putting a strain on park resources, wildlife, and infrastructure, while also degrading the visitor experience due to overcrowding.

Data & Modeling Goals To help park managers make proactive, data-driven decisions, leadership has tasked us with delivering a predictive model within a **12-week timeline**. Our target benchmark is to achieve **at least 90% accuracy**.

As a data professional on this team, your focus will be on our engineering and modeling workflow. We will be analyzing historical recreation data alongside several multi-faceted feature sets, including:

- Weather and temperature variations
- User fee structures
- Historical and real-time traffic conditions
- Other external socio-economic or seasonal factors

What's Next Your immediate role will be helping us ingest these disparate datasets, perform initial exploratory data analysis (EDA), and begin feature engineering. I know getting your hands on the data architecture and understanding our model pipeline is your top priority right now, so I will get you access to our repositories and documentation by tomorrow morning.

Please review this high-level summary, and **let me know if you have any follow-up questions** regarding the dataset schemas, our tech stack, or the project timeline. I look forward to working closely with you on this!

Best regards,

Aquib Ali
Data Scientist
National Park Service Data Team

Email #2: Email to a new writer for NPS public relations

Dear Victoria,

My name is Aquib, one of the data professionals with the National Park Service and a member of the data team responsible for the visitation prediction project.

Welcome to the National Park Service team! As a writer for our public relations department, your storytelling is vital to communicating how the NPS protects our parks while keeping them accessible. I am reaching out to share details about a new initiative our data team is launching, providing you with the background needed to craft compelling, non-technical public narratives about our efforts.

The Challenge We Are Addressing Our parks are more popular than ever. In 2022, we welcomed roughly 312 million recreation visits—an increase of 15 million from the previous year. While we love seeing the public enjoy these beautiful spaces, heavy crowding in our top 8 most visited parks (like the Grand Canyon and Great Smoky Mountains) strains fragile natural environments, impacts wildlife, and causes frustrating overcrowding for visitors.

The Solution: Proactive Park Management To tackle this, our data team is spending the next 12 weeks building a highly accurate predictive tool. By looking at historical patterns, weather trends, and traffic data, this tool will help park managers anticipate exactly when and where the heaviest crowds will arrive.

What is likely most important to your work is how this data translates into real-world improvements for the public. Managers will use these insights to make proactive, positive changes to park operations, such as:

- Optimizing staff levels to reduce wait times and lines.
- Upgrading and renovating high-use visitor facilities before they break down.
- Improving reservation and ticketing systems to make trip planning seamless for families.

How We Can Collaborate Our ultimate goal is to preserve the parks for future generations while ensuring today's visitors have an inspiring, educational, and smooth experience. This project is a great example of the NPS using modern technology to protect nature and improve public access.

I want to make sure you have everything you need to transform this technical project into engaging articles and press materials. **Please let me know if you have any follow-up questions**, or if you'd like to schedule a brief call to dive deeper into any of these points!

Warmly,

Aquib Ali
Data Scientist
National Park Service Data Team